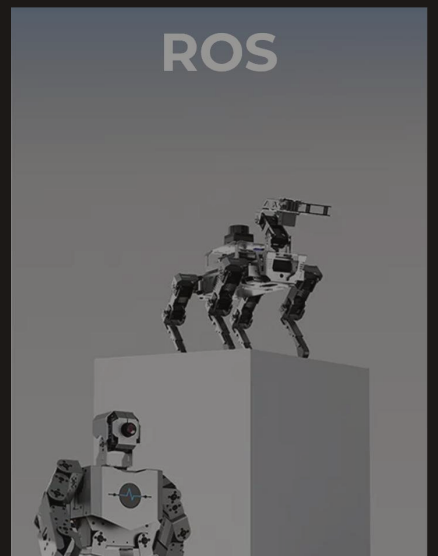
Hiwonder

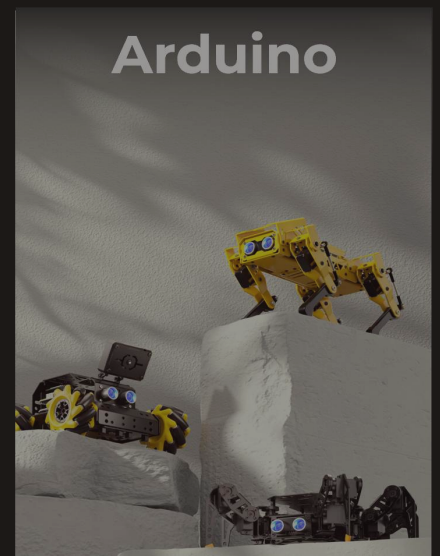
How Hiwonder Helps Students Apply AI Agents on Robots

Hardware · Software · Curriculum · Open Source · Innovation











Agenda

01 **About Hiwonder**

Company, mission, and product matrix

02 **WonderAgent**

Natural language coding for robots

03 **ESP-Claw AI Agent**

Edge AI on a hexapod platform

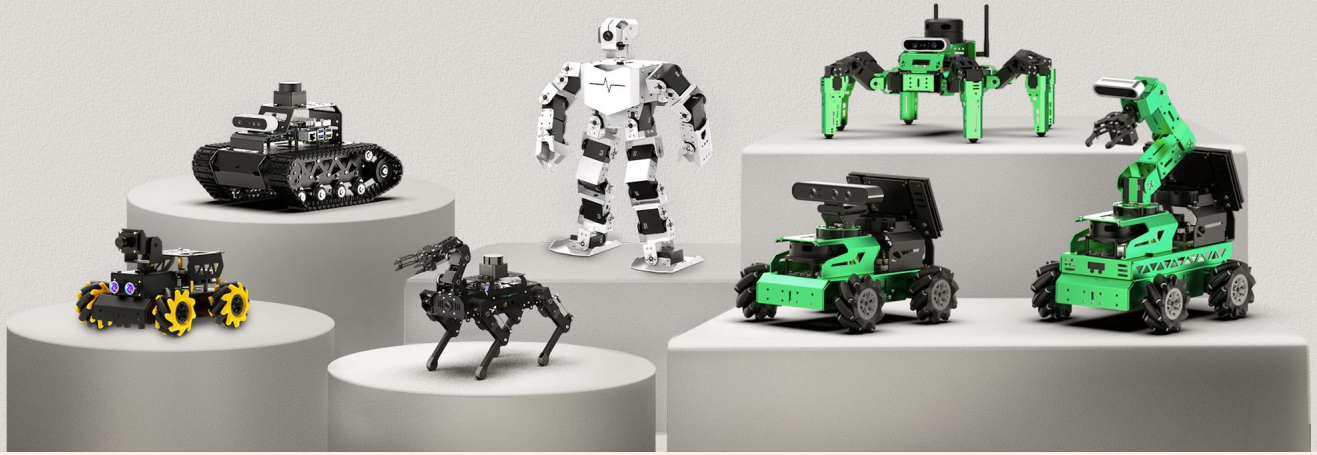
04 **ROSOrin Pro AI Agent Pipeline**

University robot car kit

05 **WonderVLA**

Imitation learning for robot arms

About Us



Our mission on education

Since 2015, We build open-source AI robots to lower the barrier to AI robotic for student
— accessible prices and a full curriculum

Factory + own servos & algorithms

Own factory. Own engineers. They develop the servos and the algorithms that move the robot.

A wide product matrix

— From K12 to university
— micro:bit, ESP32, Raspberry Pi and Jetson
— cars, arms, humanoids, quadrupeds and hexapods.

What do we provide for student ?

01

Good quality hardware

Servos, expansion board, Chassis, LiDAR……

02

Wiki on every robot

Basic operation for all function : mapping, navigate, arm, voice, vision

03

Basic course

Linux, Python, ROS 2, OpenCV

04

Competition

Meet the practical problems: light, noise, a distorted map, a slipping grasp

05

Web Tool WonderAgent

Natural Language Coding for Robots

06

Simulation

One class, few cars
Stimulate different scene
Less likely to break

Hiwonder Forum: the AI Agent answers first

01

Student posts

A question with product model, steps tried and photos

02

Agent searches

Official wiki, manuals and source code for that exact product

03

Agent replies

Step-by-step fix in minutes; asks for missing details

04

Engineer follows up

Not sure? It hands the topic to a support engineer

3.5 min

median time to first reply

≈ 50%

of questions closed with no engineer involved

24 / 7

in English and Chinese, two forums

Every answer is grounded in Hiwonder's own documentation, so students learn the official procedure — and engineers spend their time on the hard cases.

Figures: forum.hiwonder.com.cn, September 2026. English forum: forum.hiwonder.com

Real case: ROSOrin Pro won't boot — solved by the Agent

THE QUESTION

ROSOrin with a Jetson Orin NX powers on, but the Jetson never boots and no WiFi hotspot appears.

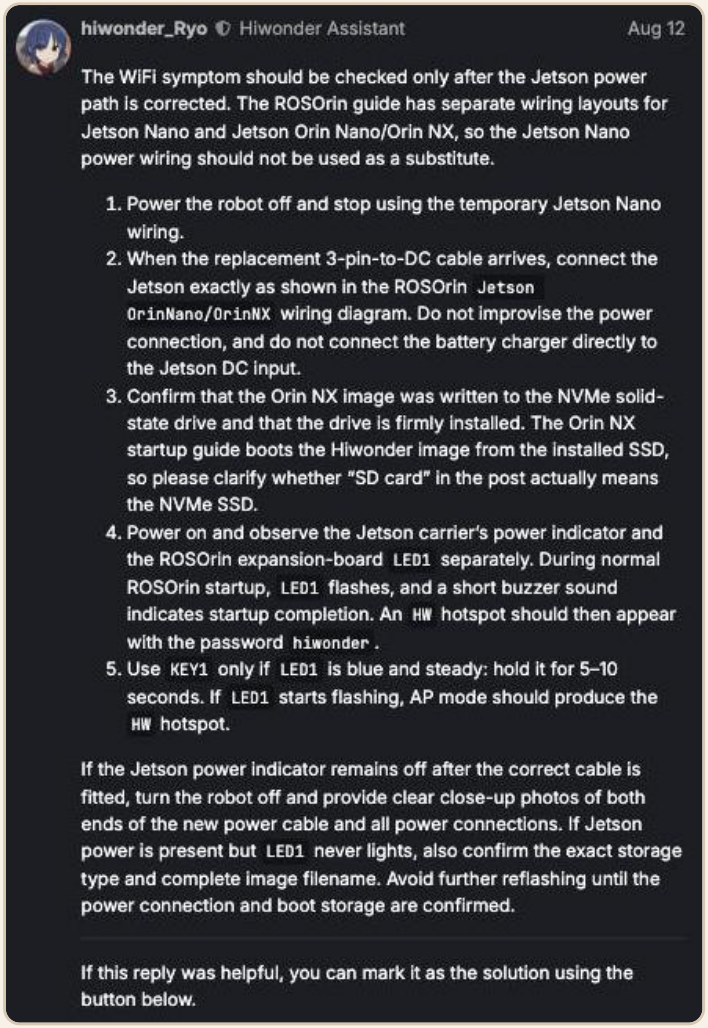
WHAT THE AGENT FOUND

- Power was wired from the Jetson Nano tutorial, not the Orin NX one
- The image was flashed to a microSD card — Orin NX boots from the NVMe SSD

RESULT

The student marked the Agent's reply as the Solution.

forum.hiwonder.com/t/238



hiwonder_Ryo Hiwonder Assistant Aug 12

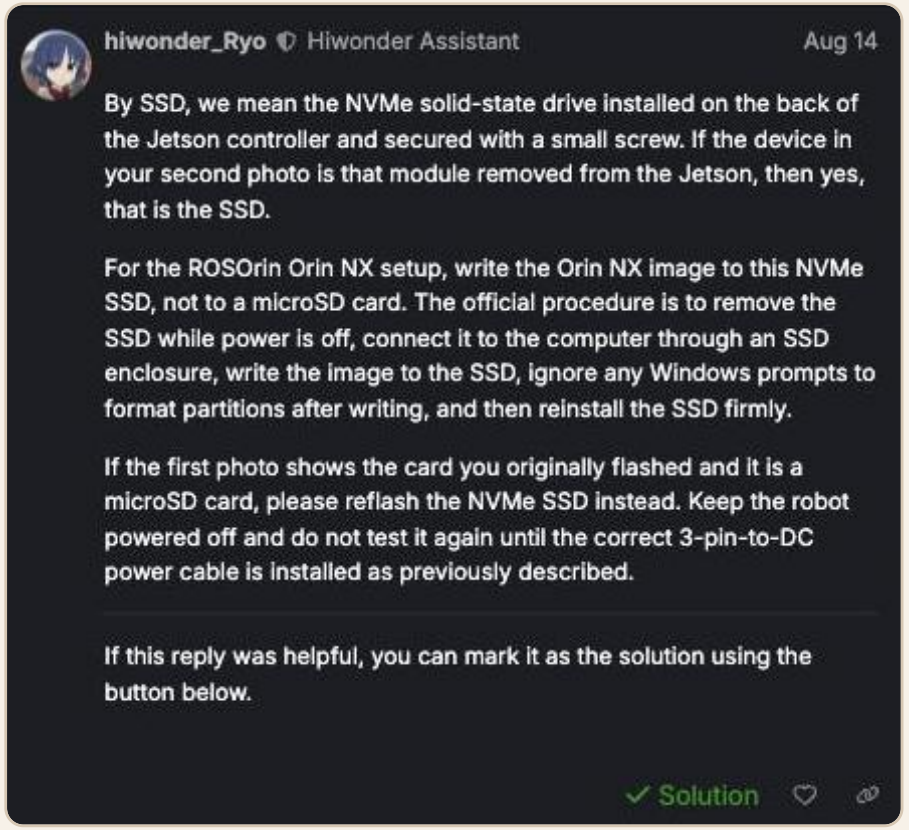
The WiFi symptom should be checked only after the Jetson power path is corrected. The ROSOrin guide has separate wiring layouts for Jetson Nano and Jetson Orin Nano/Orin NX, so the Jetson Nano power wiring should not be used as a substitute.

1. Power the robot off and stop using the temporary Jetson Nano wiring.
2. When the replacement 3-pin-to-DC cable arrives, connect the Jetson exactly as shown in the ROSOrin `Jetson OrinNano/OrinNX` wiring diagram. Do not improvise the power connection, and do not connect the battery charger directly to the Jetson DC input.
3. Confirm that the Orin NX image was written to the NVMe solid-state drive and that the drive is firmly installed. The Orin NX startup guide boots the Hiwonder image from the installed SSD, so please clarify whether "SD card" in the post actually means the NVMe SSD.
4. Power on and observe the Jetson carrier's power indicator and the ROSOrin expansion-board `LED1` separately. During normal ROSOrin startup, `LED1` flashes, and a short buzzer sound indicates startup completion. An `HW` hotspot should then appear with the password `hiwonder`.
5. Use `KEY1` only if `LED1` is blue and steady: hold it for 5–10 seconds. If `LED1` starts flashing, AP mode should produce the `HW` hotspot.

If the Jetson power indicator remains off after the correct cable is fitted, turn the robot off and provide clear close-up photos of both ends of the new power cable and all power connections. If Jetson power is present but `LED1` never lights, also confirm the exact storage type and complete image filename. Avoid further reflashing until the power connection and boot storage are confirmed.

If this reply was helpful, you can mark it as the solution using the button below.

Reply 1: fix the power path, check the boot drive



hiwonder_Ryo Hiwonder Assistant Aug 14

By SSD, we mean the NVMe solid-state drive installed on the back of the Jetson controller and secured with a small screw. If the device in your second photo is that module removed from the Jetson, then yes, that is the SSD.

For the ROSOrin Orin NX setup, write the Orin NX image to this NVMe SSD, not to a microSD card. The official procedure is to remove the SSD while power is off, connect it to the computer through an SSD enclosure, write the image to the SSD, ignore any Windows prompts to format partitions after writing, and then reinstall the SSD firmly.

If the first photo shows the card you originally flashed and it is a microSD card, please reflash the NVMe SSD instead. Keep the robot powered off and do not test it again until the correct 3-pin-to-DC power cable is installed as previously described.

If this reply was helpful, you can mark it as the solution using the button below.

✓ Solution

Reply 2 (marked **Solution**): flash the NVMe SSD

03 WonderAgent

Natural Language Development Tool for Embedded board

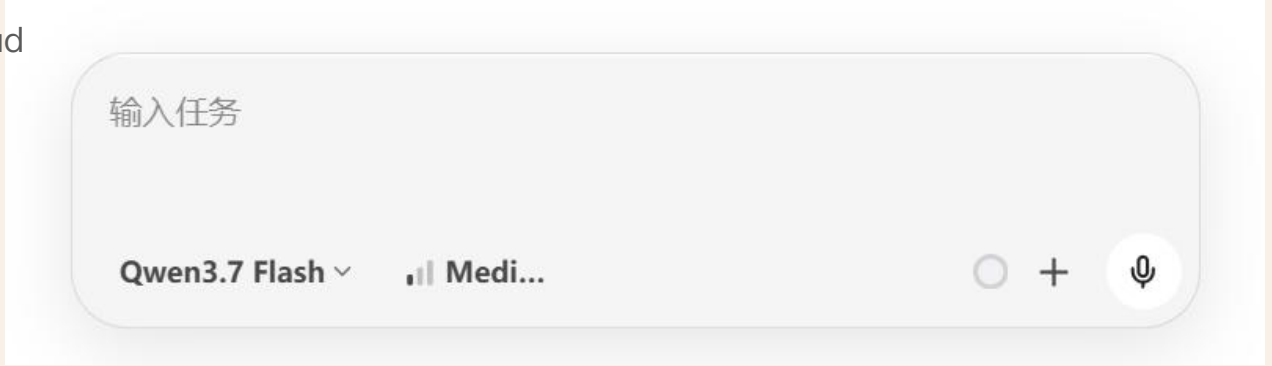


1 Describe

Tell WonderAgent what you want in plain language

2 Generate

It writes the code and compiles the firmware in the cloud



3 Download

Connect the product and flash in one click

https://wonderagent.hiwonder.com.cn/

From one sentence to a running function

01 **Describe**

Natural language in the chat

02 **Generate**

Code + cloud compile

03 **Download**

Cable → flash firmware

04 **Experience**

Product runs immediately

Make firmware development as simple as chatting

LIVE DEMO

The entire process happens in one webpage.
No local toolchain installation required.

What else can WonderAgent do ?

✅ What I can do

category	specific abilities
Code writing	Read, create, and edit `.ino` files `.cpp`, `.h` implementing new logic according to the hardware manual and existing APIs.
Compile and build	Use `build` tools to compile Arduino/C++ projects and obtain complete compiler errors for debugging.
firmware flashing	Use `flash` the tool to write the compiled firmware to the connected development board (you need to click "Connect Device" and select the serial port first).
File search	Full text search (`search_files`), filename matching search (`find_files`), directory browsing (`list_directory`)
Document search	Search official product catalogs/chip information, search public document summaries
Workshop Browsing	View approved works from the creative workshops, as well as your own submissions.
Engineering Analysis	Understand the existing driver layer (such as `HW_MechDog`, `pwm_servo`, `action.h`) and design upper-layer applications based on it.

i want to know more about the Parameter Description

I have been thinking for 20 seconds ›

Okay, let me delve into the parameter descriptions for each level of the project .

📐 MechDog AI Complete Parameter Manual

I. Gait parameters `set_gait_params()`

Invocation method:

```cpp
mechdog.set_gait_params(swingTime, supportTime, height);
```

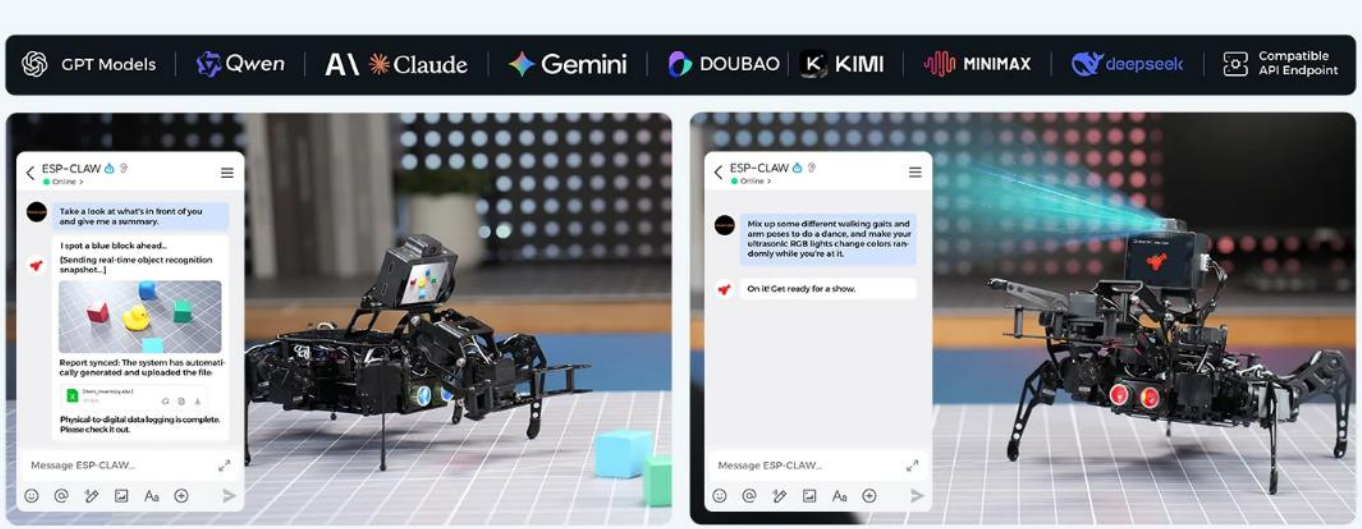
parameter	type	default value	meaning
swingTime (swing time)	`int` ms	150	Duration of each foot off the ground
supportTime	`int` ms	200	Duration of each foot on the ground
height (leg lift height)	`int` mm	25	Maximum height of feet off the ground

Recommended parameter range:

Scene	swingTime	supportTime	height	Effect
Normal walking	150	200	25	Default Balanced Attitude
walk fast	150	350	20	Increase frequency
Steady progress	200	600	50	Slow but rising, steady
Color Follow	150	450	40	Flexible steering
Interval running	150	200	25	Small steps, fast frequency

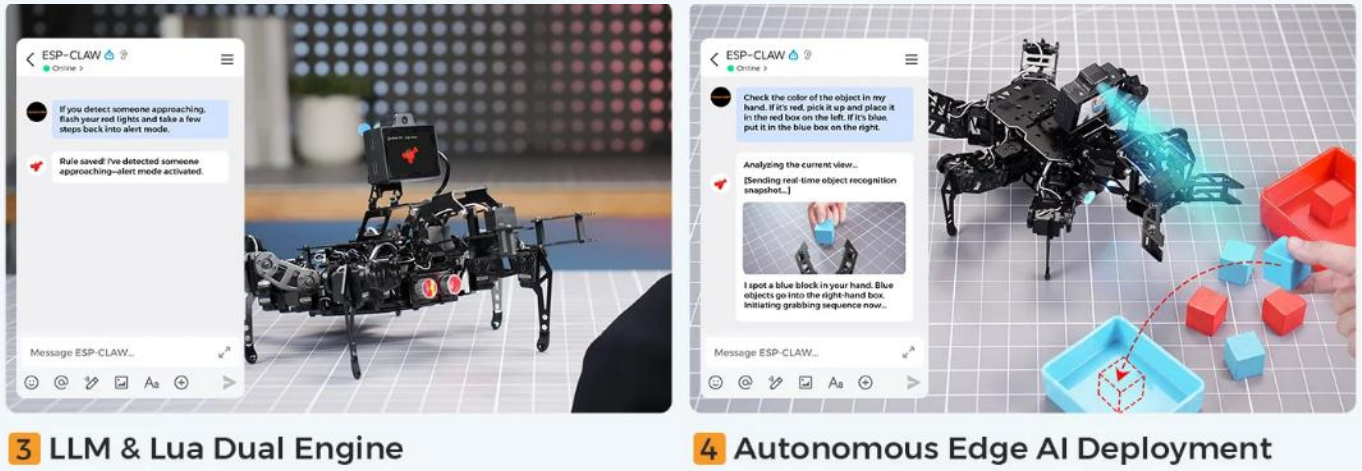
> ⚠Note : This only affects **the next time** it is called . When switching speeds, call it first , then ...
> `move()` `set_gait_params()` `move()`

Built-in ESP-Claw AI Agent



1 Cross-Platform Collaboration

2 No-Code Voice Programming



3 LLM & Lua Dual Engine

4 Autonomous Edge AI Deployment

ROSOrin Pro —Robot Car Kit for University

SEE



3D depth camera

Supports YOLO & OPENCV
Color, object, face, 3d pose

WHERE



TOF LiDAR

SLAM mapping and Nav2 navigation with live obstacle avoidance

BRAIN



Jetson + STM32

Jetson — GPU for onboard AI
STM32 — real-time control of motors and servos



HEAR



AI voice module

Wake word, voice command, Voice interaction

DO



6-DOF arm

Grasping or planned motion

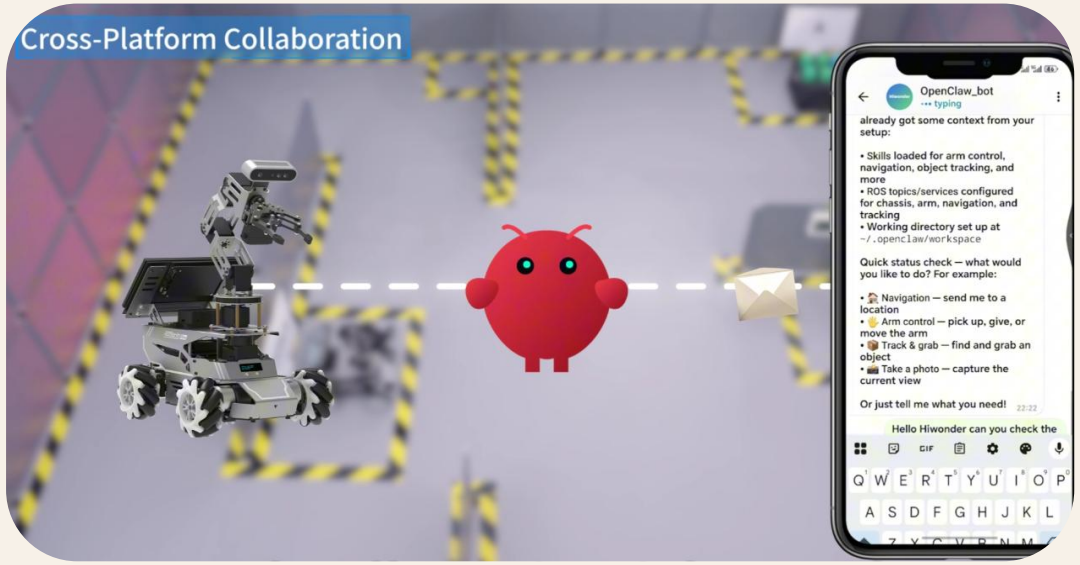
MOVE



Metal Mecanum chassis

Encoders + IMU
Support different chassis for different project and terrain

How OpenClaw is applied on ROSOrin Pro



01

You speak or type

TUI, app, or WonderEcho wake word + ASR.

02

OpenClaw decides

LLM splits the sentence into skills: move, look, go, grasp.

03

ROS nodes run

Topics and services: /cmd_vel, arm group, Nav2 goal, camera.

04

The body acts

Jetson thinks. STM32 moves motors and servos. Agent replies in text.

Move

Chassis Twist on /cmd_vel

See

Camera + color / depth track

Go

Nav2 goal on the SLAM map

Grasp

Arm action + IK / servos

Reply

LLM text back to the TUI

HOST · Jetson Orin NX — High-level stack · sensors attach over USB

- **Depth camera** — USB · RGB + depth
- **LiDAR** — USB · 2D scan
- **WonderEcho Pro** — serial ASR

On-board compute

SLAM · YOLO / TensorRT · Nav2 · OpenClaw gateway · Wi-Fi

◄ **USB-UART /dev/nrc · 1 Mbps · RRC protocol** ►

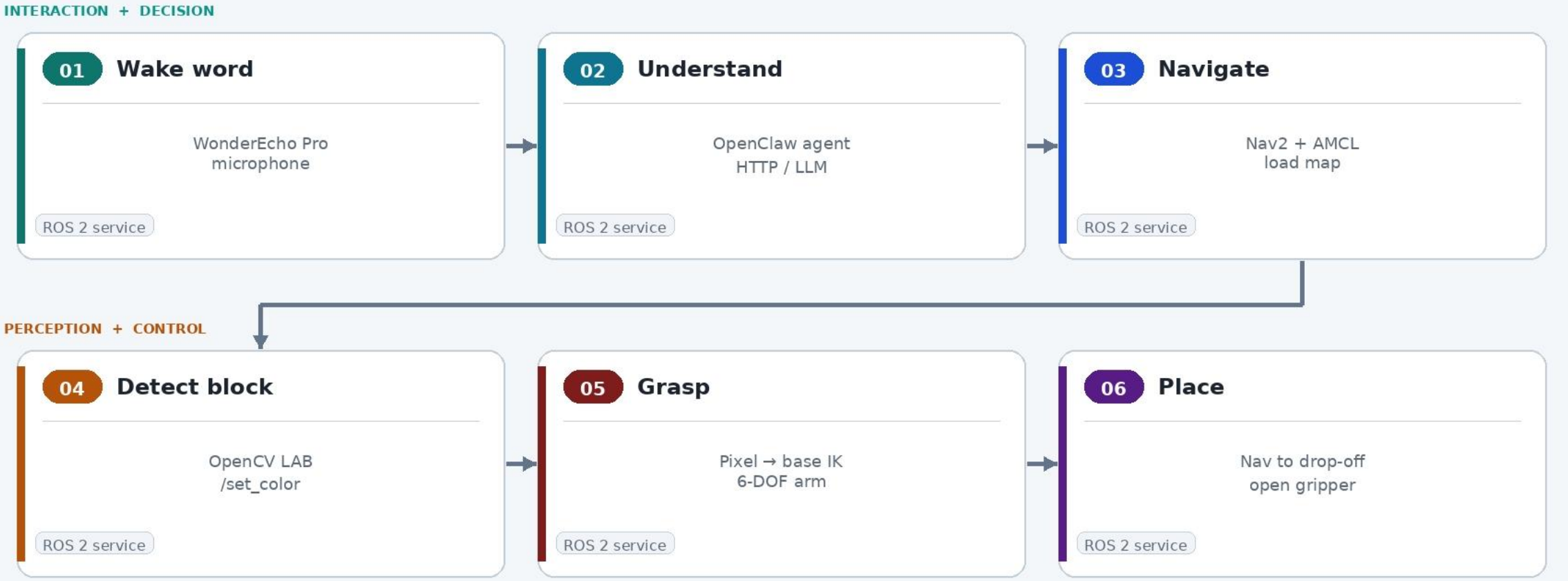
host → STM32 commands down · encoder / IMU / battery up ←

MCU · STM32 — Real-time loop · actuators and onboard sense

- **Chassis motors** — PWM · Mecanum drive
- **6-DOF arm** — bus servos HX-12H
- **OLED + IMU** — I2C / SPI
- **Battery ADC** — onboard sense

Real-time loop stays on the MCU · YOLO / SLAM load cannot stall PWM or servo bus updates

How ROSOrin Pro Uses OpenClaw and ROS 2 for Robot Tasks



Example agent scenarios

Fruit *“I ordered fruit. Pickup code 1188.”*

Nav2 → vision confirms label → grasp → home

Package *“Also grab tracking number 311W-WBD01G*

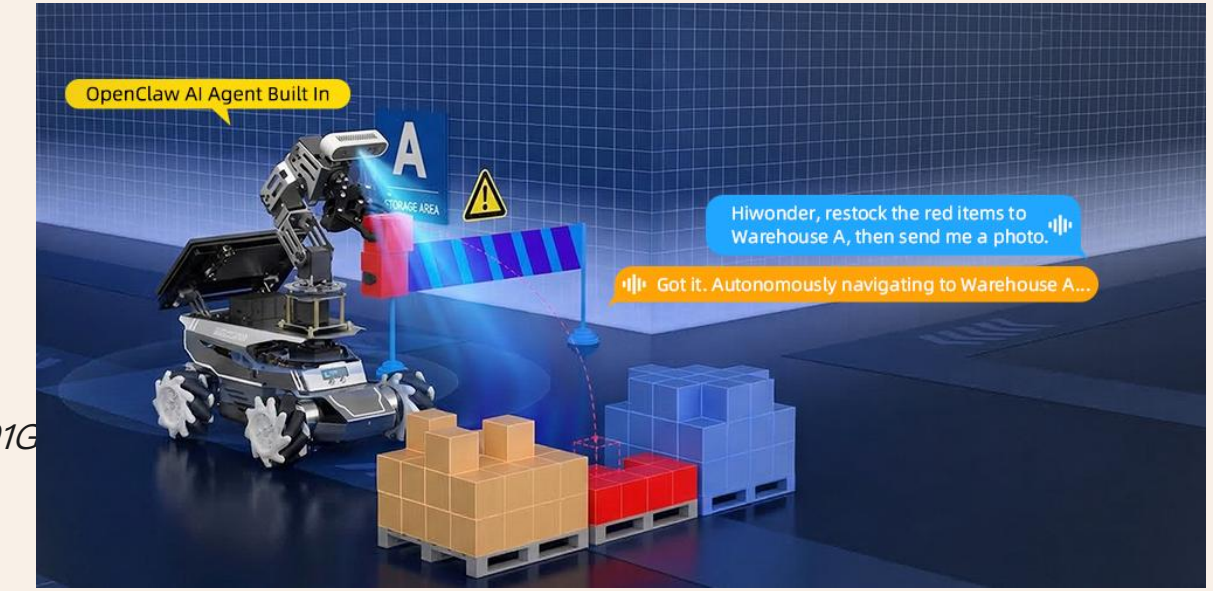
Keep context → replan → read label → grasp

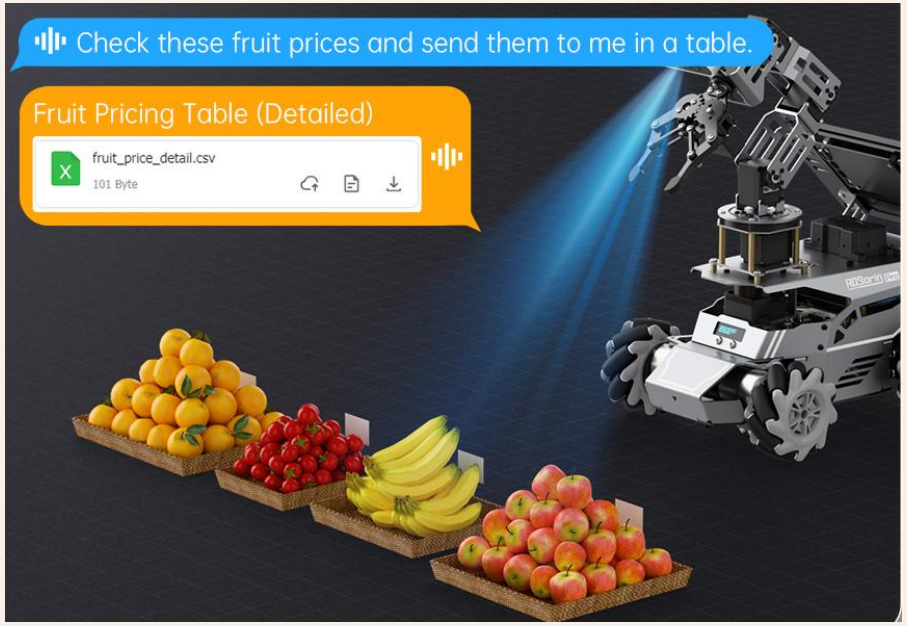
Find dog *“Find the dog and send me a photo.”*

Switch class → patrol → photograph → send

Home *“Head home.”*

Mark jobs done → return → report





05 WonderVLA

Imitation Learning Workbench for Robot Arms

1 Record

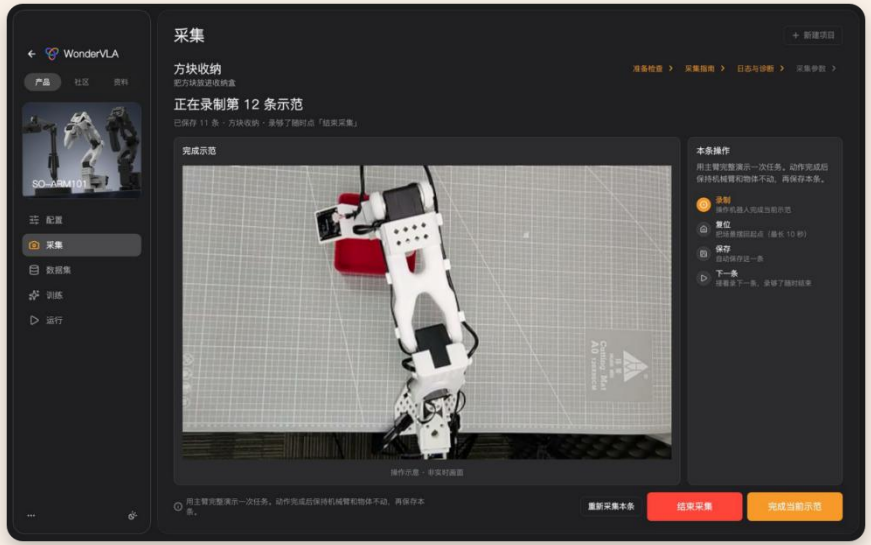
Move the leader arm by hand; the follower and two cameras record each demonstration

2 Train

Train an ACT or SmolVLA policy on your own GPU or a cloud server

3 Run

The arm repeats the task on its own, then parks safely





Desktop app for macOS · Windows · Linux | works with SO–ARM101

From a few demonstrations to an autonomous arm

01

Set up

Ports, cameras, auto-calibration

02

Record

Teleoperate, 30—50 episodes

03

Train

ACT / SmolVLA, local GPU or cloud

04

Run

The arm does the task by itself

LIVE DEMO

The whole LeRobot workflow in one desktop app, guided step by step.
Python runtime, drivers and models are installed for you.

Thank You

Hiwonder

https://www.hiwonder.com